

Agenda

1. Friday Feedback →
2. Section 7.5
3. Section 7.8



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Section 7.5 — Guidelines for Integration

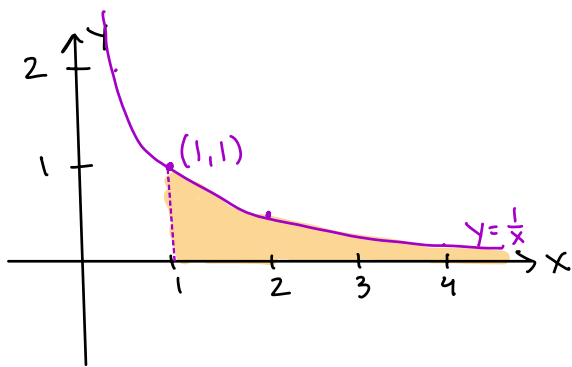
1. Simplify the integrand if possible
2. Look for an obvious substitution
3. Classify the integrand according to its form:
 - (a) Trigonometric Functions
 - (b.) Rational Function
 - (c.) Products
 - (d.) Radicals
4. Try Again

Extra Practice:

- ①. $\int x^2 \sin^2(x^3) dx$ $u = x^3$
②. $\int (5x-1) \cos x dx$ IBP $u = 5x-1 \quad dv = \cos x dx$
③. $\int \frac{x^2}{(4+x^2)^{3/2}} dx$ Trig sub
- ④. $\int \frac{dx}{\sqrt{4x^2+25}}$ Trig sub
⑤. $\int_0^1 \frac{dx}{(x+1)(x^2+9)}$ P. Frac.
⑥. $\int x^3 \ln x dx$ IBP
- ⑦. $\int \tan x \sec^3 x dx$ Trig funct.
⑧. $\int \tan^2 x \sec^4 x dx$ Trig funct.
⑨. $\int \frac{dx}{x\sqrt{x^2-16}}$ Trig sub.
- ⑩. $\int \frac{x+3}{x(x^2+1)^2} dx$ P. Frac.
⑪. $\int \sin(5x) \cos(2x) dx$ Trig funct. (product identity)
⑫. $\int (2x^3+1) \sin(x^4+2x) dx$ Vsub

Section 7.8 - Improper Integrals

Motivation:



Let $t > 1$

$$\textcircled{1} \int_1^t \frac{1}{x} dx =$$

$$\ln|x| \Big|_1^t$$

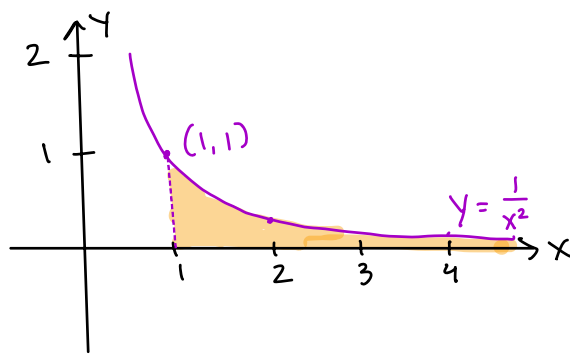
$$= \textcircled{\ln t}$$

what happens as $t \rightarrow \infty$

$$\textcircled{1} \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x} dx$$

$$= \lim_{t \rightarrow \infty} \ln t = \textcircled{\infty}$$

$\int_1^{\infty} \frac{1}{x} dx$ diverges



$$\textcircled{2} \int_1^t \frac{1}{x^2} dx =$$

$$-x^{-1} \Big|_1^t$$

$$= \textcircled{1 - 1/t}$$

$$\textcircled{2} \lim_{t \rightarrow \infty} \int_1^{\infty} \frac{1}{x^2} dx$$

$$\lim_{t \rightarrow \infty} \left[1 - \frac{1}{t} \right] = 1$$

$\int_1^{\infty} \frac{1}{x^2} dx$ converges

Improper Integrals on Infinite Integrals

Definitions: Let a be a constant.

1.) If $\int_a^t f(x) dx$ exists for all $t \geq a$, then we define

$$\int_a^{\infty} f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx$$

2.) If $\int_t^a f(x) dx$ exists for all $t \leq a$, then we define

$$\int_{-\infty}^a f(x) dx = \lim_{t \rightarrow -\infty} \int_t^a f(x) dx$$

We say that the improper integral converges if it exists, and we say it diverges if it does not exist.

Activity: $\int_0^{\infty} x e^{-x^2} dx$
u-sub

$$\lim_{t \rightarrow \infty} \int_0^t x e^{-x^2} dx \quad \begin{array}{l} u = -x^2 \\ du = -2x dx \end{array}$$

$$-\frac{1}{2} e^u du \Big|_0^{t^2}$$

$$\lim_{t \rightarrow \infty} \left[-\frac{1}{2} e^{-t^2} + \frac{1}{2} \right]$$

$$\hookrightarrow \lim_{t \rightarrow \infty} \left[-\frac{1}{2} e^{-t^2} + \frac{1}{2} \right] = \frac{1}{2}, \text{ converges}$$

If both $\int_0^{\infty} f(x) dx$ and $\int_{-\infty}^0 f(x) dx$ converge, then

$$\int_{-\infty}^{\infty} f(x) dx = \int_0^{\infty} f(x) dx + \int_{-\infty}^0 f(x) dx$$

Example $\int_{-\infty}^{\infty} \frac{dx}{\pi(1+x^2)}$

$$\int_0^{\infty} \frac{dx}{\pi(1+x^2)} = \lim_{t \rightarrow \infty} \frac{1}{\pi} \int_0^t \frac{dx}{1+x^2} = \lim_{t \rightarrow \infty} \frac{1}{\pi} \arctan(x) \Big|_0^t = \frac{1}{2}$$

$$\int_{-\infty}^0 \frac{dx}{\pi(1+x^2)} = \lim_{t \rightarrow -\infty} \frac{1}{\pi} \int \frac{dx}{1+x^2} = \lim_{t \rightarrow -\infty} \frac{1}{\pi} \arctan(x) \Big|_0^t = \frac{1}{2}$$

$$\int_{-\infty}^{\infty} \frac{dx}{\pi(1+x^2)} \text{ converges to } \frac{1}{2} + \frac{1}{2} = \textcircled{1}$$

Activity: $\int_{-\infty}^{\infty} x dx$

$$\int_0^{\infty} x dx = \lim_{t \rightarrow \infty} \int_0^t x dx = \lim_{t \rightarrow \infty} \left. \frac{1}{2} x^2 \right|_0^t$$

$$= \lim_{t \rightarrow \infty} \frac{1}{2} t^2 = \infty$$

(diverges)

Both $\int_0^{\infty} f(x) dx$ and $\int_{-\infty}^0 f(x) dx$ must converge for $\int_{-\infty}^{\infty} f(x) dx$ to converge.

p-Integrals

Theorem: Let p be a constant.

- 1.) If $p > 1$, then the improper integral $\int_1^{\infty} \frac{dx}{x^p}$ Converges.
- 2.) If $p \leq 1$, then the improper integral $\int_1^{\infty} \frac{dx}{x^p}$ diverges.

Examples: $\int_1^{\infty} \frac{dx}{x^5}$, $\int_1^{\infty} \frac{dx}{\sqrt{x}}$, $\int_1^{\infty} \frac{dx}{x^{1.0001}}$, $\int_1^{\infty} x^3 dx$